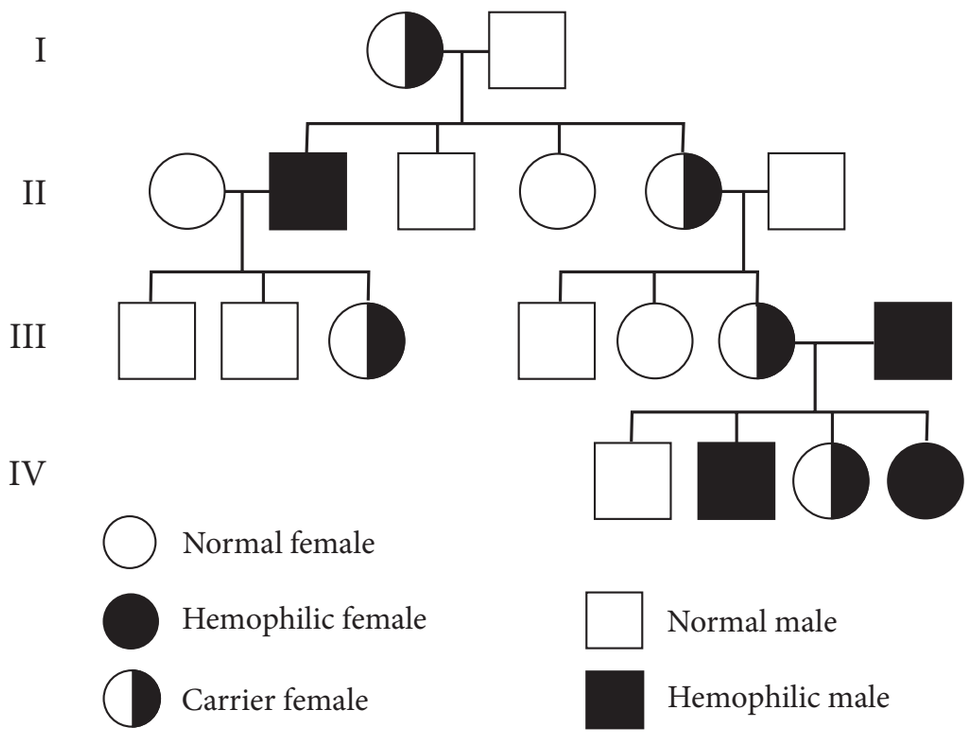
## 20. X – LINKED DISEASE - HAEMOPHILIA (BLEEDER'S DISEASE)

### IDENTIFICATION

The given pedigree chart is identified as the genetic disease Haemophilia.

### COMMENTS

1. Haemophilia or bleeder's disease (Royal disease) is the most notorious of all sex-linked diseases. The person suffering from this disease bleeds for a long period (30 minutes to 24 hours) during injury due to the failure of blood coagulation.
2. It is caused by a recessive X – linked gene more common in men than women.
3. The females are carriers of the disease and would transmit the disease to 50% of their sons even if the male parent is normal.
4. It follows criss – cross or zig – zag pattern of inheritance (i.e., grandfather transmits his X linked character to his grandson through carrier daughter).




### DISCUSSION QUESTIONS

1. Observe the given pedigree chart and identify the affected individuals and carriers in the II generation.
2. Why are men affected often in X linked inheritance?
3. What is the pattern of inheritance in the given pedigree chart?
4. Why are women said to be carriers in X linked inheritance?
5. How is haemophilia caused?



XII Std Biology-Zoology Practical EM.indd 248 23/12/2022 14:22:47